

Worksheet
Integration: Substitution / Integration by Parts / Trig Methods

1. $\int \frac{e^{2x}}{1+e^{4x}} dx$

2. $\int_0^4 \frac{dx}{(1+x)\sqrt{x}}$

3. $\int \frac{\tan x}{\cos^2 x} dx$

4. $\int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x} d\theta$

5. $\int x^2 \sin x dx$

6. $\int_0^1 \ln \sqrt{x^2+1} dx$

7. $\int x \sin x^2 dx$

8. $\int_0^{\pi} \cos^5 x dx$

9. $\int \tan^4 x \sec^2 x dx$

10. $\int_0^4 \sqrt{16-x^2} dx$

11. $\int e^x \sqrt{1+e^{2x}} dx$

12. $\int_0^1 \frac{dx}{4x^2+4x+5}$

13. $\int x^2 \sqrt{4-x^2} dx$

14. $\int_0^2 x \sqrt{4-x^2} dx$

15. $\int x \sin^2 x dx$

16. $\int_0^1 x \sin^{-1} x dx$